

Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	Smart agriculture production optimization engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
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**1 Frontend /
Client-Side**

*[Explain why this was
chosen (e.g., responsive
design, component
reusability)]*

*(e.g., React, Angular,
HTML/CSS)
[Specify
technology]*

2	Backend / Server-Side <i>[Specify Technology]</i> <i>PYTHON</i> <i>(e.g., Node.js, Python/Django, Java)</i>	<i>[Explain why this was chosen (e.g., fast processing, existing team expertise)]</i>
3	Database / Data Storage <i>[Specify Technology]</i> <i>SQL</i> <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>[Explain why this was chosen (e.g., handles unstructured data, strong relational mapping)]</i>
4	Cloud / Hosting / Deployment <i>[Specify technology]</i> <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>[Explain why this was chosen (e.g., easy scalability, cost-effective hosting)]</i>

5	Version Control & CI/CD <i>[Specify technology]</i>	<i>[Explain why this was chosen (e.g., seamless collaboration, automated testing)]</i>
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	<i>(e.g., GitHub, GitLab, Jenkins)</i>	
6	Third-Party APIs / Other Tools <i>[Specify technology]</i> <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>[Explain the specific feature this enables in your project]</i>